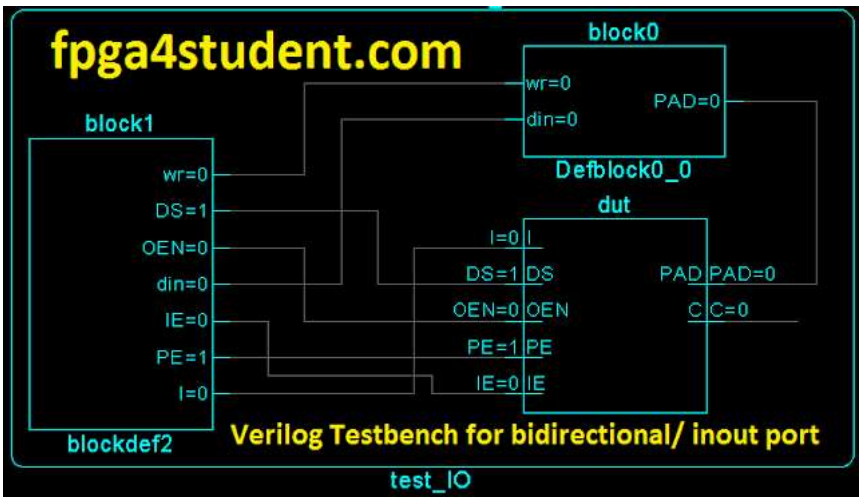
# How to write Verilog Testbench for bidirectional/ inout ports

**This post describes how to write a Verilog testbench for bidirectional or inout ports. This happens in special designs which contain bidirectional or inout ports such as I2C core, IO pads, memories, etc.**

**In this post, I will give an example how to write testbench code for a digital IO pad. Basically, the IO pad has logic inputs DS, OEN, IE, PE to configure the IO pad as an input or output. When DS = OEN = IE = PE = 1, the IO pad operates as an input pad. Thus, the data from the bidirectional port PAD are written into the output C. When DS = PE =1 and OEN = IE = 0, the IO pad operates as an output pad. Therefore, the signal from the input I is passed to the bidirectional port PAD.**




Block diagram of the Verilog testbench

**As shown in the block diagram of the Verilog testbench, the IO pad under test is the DUT block. Block 0 controls the inout port PAD as follows:**

**assign PAD = wr==1 ? din:1'bz;**

**When wr==1, the inout port PAD operates as an input, so PAD = din. When wr==0, the inout port PAD is assigned to a high impedance 1'bz. Thus, it becomes an output to get data from the input I of the IO pad.**

**Block 1 is the main testbench to generate test cases to check the input/ output functionality of the IO pad.**

**Below is the full Verilog testbench for the IO pad.**

```
`timescale 1ns/10ps
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module test_IO();

reg DS,OEN,IE,PE,I,din;
wire PAD;
reg wr;
wire C;
// inout port
assign PAD = wr==1 ? din:1'bz;

DIGITAL_IO dut(I,DS,OEN,PAD,C,PE,IE);

initial begin
 // test the pad as an input
 wr=1;
 DS=1;
```
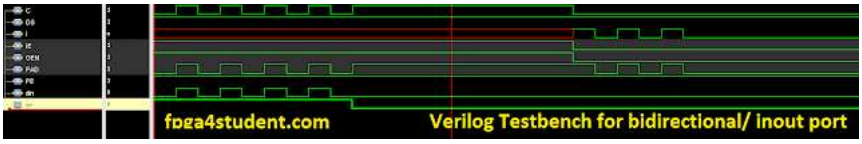

```
 OEN=1;
 din=0;
 IE=1;
 PE=1;
 #100;
 din=1;
 #100;
 din=0;
 #100;
 din=1;
 #100;
 din=0;
 #100;
 din=1;
 #100;
 din=0;
 #100;
 din=1;
 #100;
 din=0;
 #100;
 wr=0;
 // test the pad as an output
 // by reading data out
 #1000;
 DS=1;
 OEN=0;
 din=0;
 IE=0;
 PE=1;
 I=1;
 #100;
 I=0;
 #100;
 I=1;
 #100;
 I=0;
 #100;
 I=1;
 #100;
 I=0;
 #100;
 end

endmodule
```

**Simulation waveform for the bidirectional port:**




**The simulation waveform shows that when wr= DS = OEN = IE = PE = 1, the IO pad operates as an input pad and the signal from bidirectional port PAD is passed to the output C. When DS = PE =1 and OEN = IE = 0, wr=0 enables the IO pad operating as an output pad. Therefore, the signal from the bidirectional port PAD is read from the input port I.**
**In short, the takeaway stuff is the following 'assign' Verilog statement:**

**assign PAD = (wr==1) ? din:1'bz;**

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